TH450/TH550·TH450A/TH550A

Industrial Robot

Manual for TH450/TH550 and TH450A/TH550A Differences

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Shibaura Machine

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Introduction

This manual describes the differences between the new SCARA robots TH450A/TH550A and the previous models TH450/TH550.
This manual is divided into the sections below.

Chapter 1: Specifications

Chapter 2: Robot Height and Z-axis Position

Chapter 3: Cables Between Robot and Controller

Chapter 4: HAND Input Signal Layout and Number of Inputs/Outputs

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1. Specifications

[Comparison of TH450A and TH450 Specifications]

Compared to the TH450, the TH450A has improved standard cycle time and permissible load inertia.

Additionally, combined with the TS3000 controller, the TH450A has increased hand wires.

No.	Name		TH450	**TH450A**	Remarks
No.	Item		Specification		
1	Type		Horizontal multi-articulation		
2	No. of controlled axes		4 axes		
3	Arm length	Full length	450 (mm)		
		Arm 1	200 (mm)		
		Arm 2	250 (mm)		
4	Working envelope	Axis 1	±120 (deg)		
		Axis 2	±145 (deg)		
		Axis 3	0 to 150 (mm)		
		Axis 4	±360 (deg)		
5	Maximum speed	Axis 1	600 (deg/sec)		
		Axis 2	600 (deg/sec)		
		Axis 3	2000 (mm/sec)		
		Axis 4	2000 (deg/sec)		
		Composite	7.3 (m/sec)		
6	Payload mass	Rated	2 (kg)		*1
		Max	5 (kg)		
7	Standard cycle time (when transferring 2 kg)		0.33 (sec)	**0.30 (sec)**	*2
8	Permissible load inertia		0.05 (kg•m^2)	**0.06 (kg•m^2)**	*1
9	Positioning repeatability	X-Y	±0.01 (mm)		*3
		Z (axis 3)	±0.01 (mm)		
		C	±0.005 (deg)		
10	Drive system		AC servo motor for all axes		
11	Robot body	Mass	27 kg	**26 kg**	
12	Hand wires and pipes		5 inputs and 4 outputs 4 mm diameter x 4	**8 inputs and 8 outputs** **4 mm diameter x 4**	
13	Options		• Ceiling suspension model • Z-axis long stroke • Top cap model	• Ceiling suspension model • Z-axis long stroke • Top cap model • **IP** • **Cleanroom model**	*4

*1 : Maximum speed rates and acceleration rates are limited depending on motion patterns, payload mass, and load inertia.

*2: Continuous operation of standard cycle motion pattern is not possible beyond the effective load ratio.
(Horizontal 300 mm, vertical 25 mm, round-trip, coarse positioning)

*3: When the environment temperature is constant.

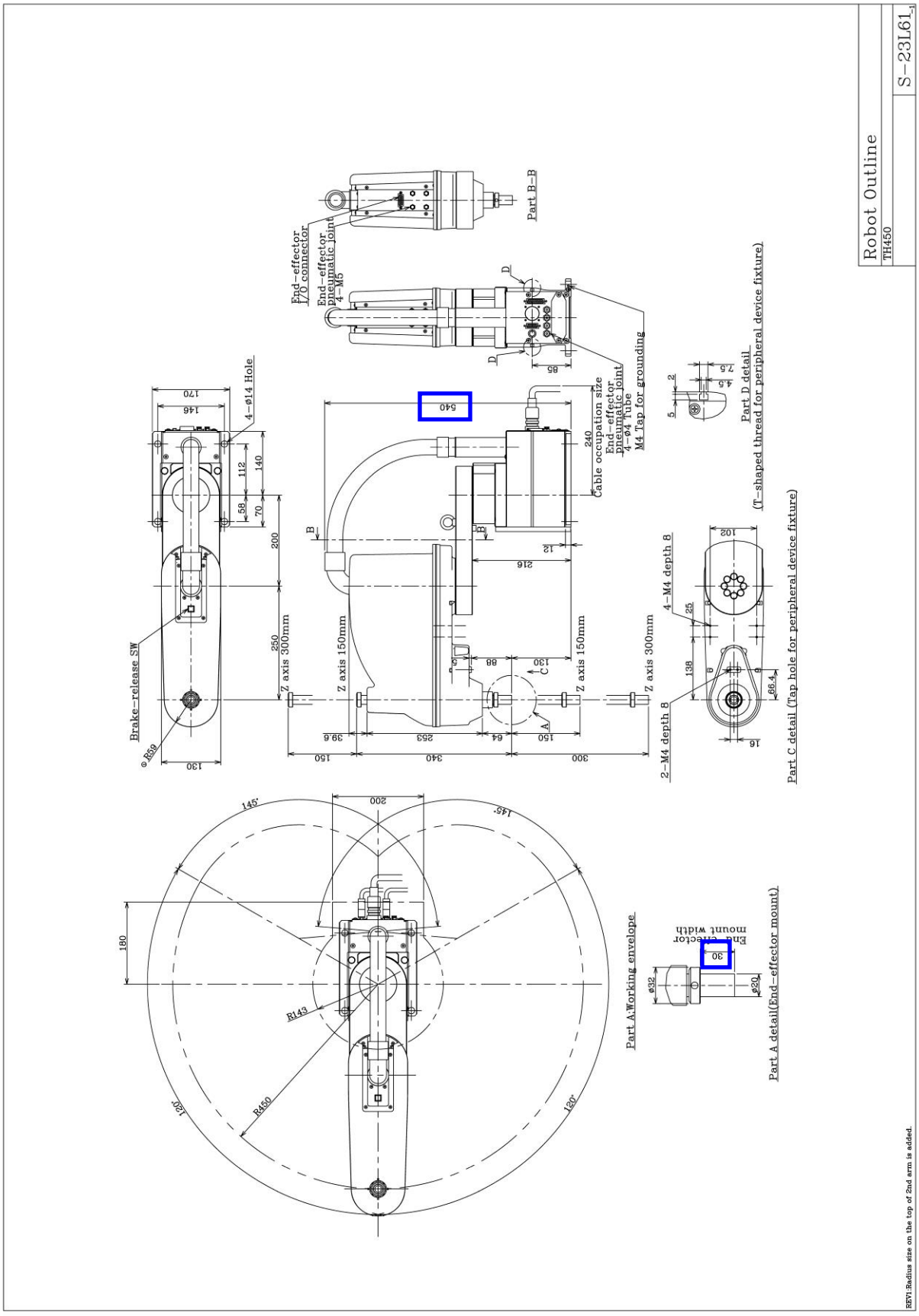
*4 : For information about the availability of other options, please contact the Sales Department.

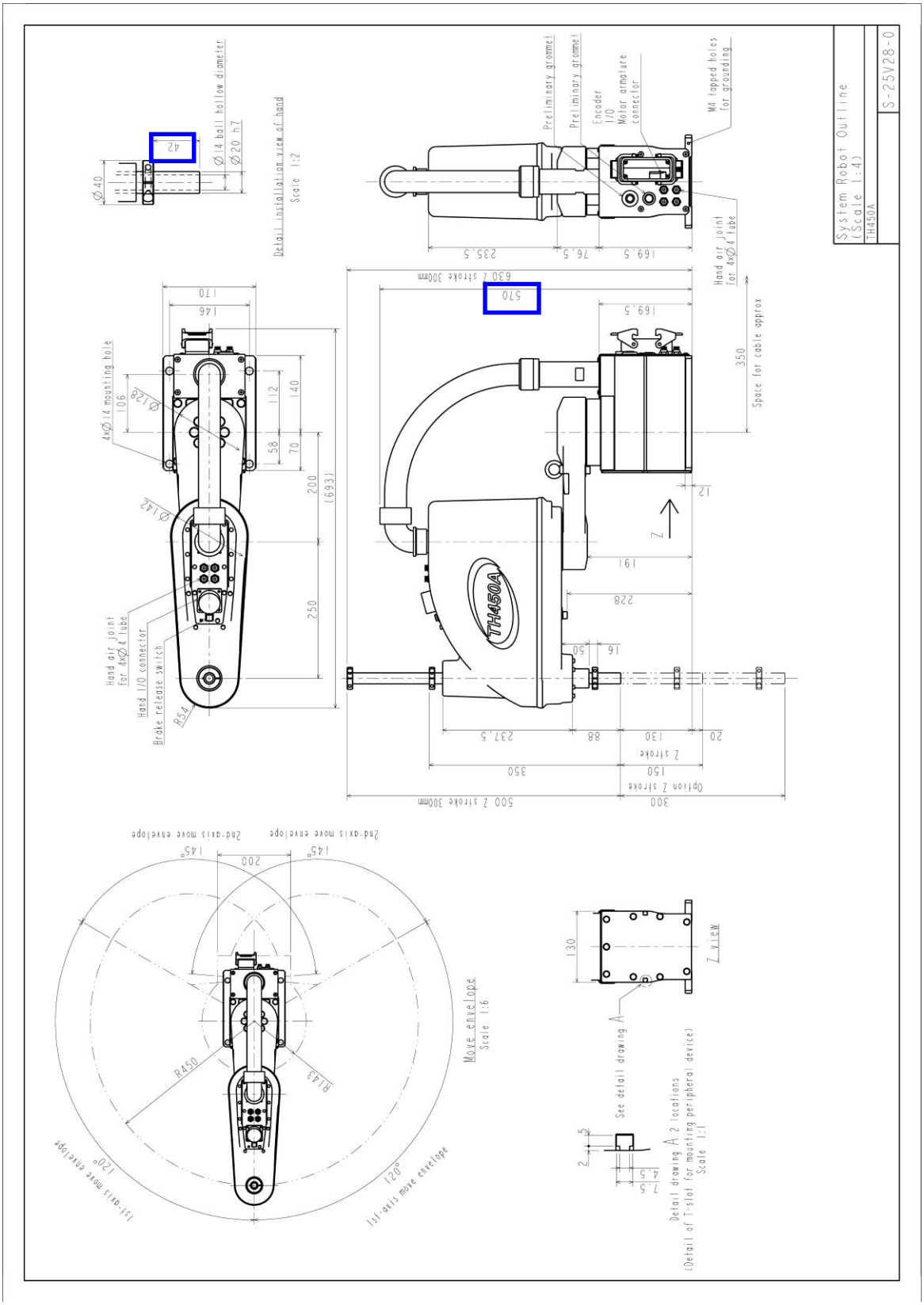
[Comparison of TH550A and TH550 Specifications]
Compared to the TH550, the TH550A has improved standard cycle time and permissible load inertia.
Additionally, combined with the TS3000 controller, the TH550A has increased hand wires.

	Name		TH550	**TH550A**	Remarks
No.	Item		Specification		
1	Type		Horizontal multi-articulation		
2	No. of controlled axes		4 axes		
3	Arm length	Full length	550 (mm)		
		Arm 1	300 (mm)		
		Arm 2	250 (mm)		
4	Working envelope	Axis 1	±120 (deg)		
		Axis 2	±145 (deg)		
		Axis 3	0 to 150 (mm)		
		Axis 4	±360 (deg)		
5	Maximum speed	Axis 1	375 (deg/sec)		
		Axis 2	600 (deg/sec)		
		Axis 3	2000 (mm/sec)		
		Axis 4	2000 (deg/sec)		
		Composite	6.21 (m/sec)		
6	Payload mass	Rated	2 kg		*1
		Max	5 kg		
7	Standard cycle time (when transferring 2 kg)		0.33 (sec)	**0.30 (sec)**	*2
8	Permissible load inertia		0.05 (kg•m^2)	**0.06 (kg•m^2)**	*1
9	Positioning repeatability	X-Y	±0.01 (mm)		*3
		Z (axis 3)	±0.01 (mm)		
		C	±0.005 (deg)		
10	Drive system		AC servo motor for all axes		
11	Robot body	Mass	29 kg	**28 kg**	
12	Hand wires and pipes		5 inputs and 4 outputs 4 mm diameter x 4	**8 inputs and 8 outputs** **4 mm diameter x 4**	
13	Options		• Ceiling suspension model • Z-axis long stroke • Top cap model	• Ceiling suspension model • Z-axis long stroke • Top cap model • **IP** • **Cleanroom model**	*4

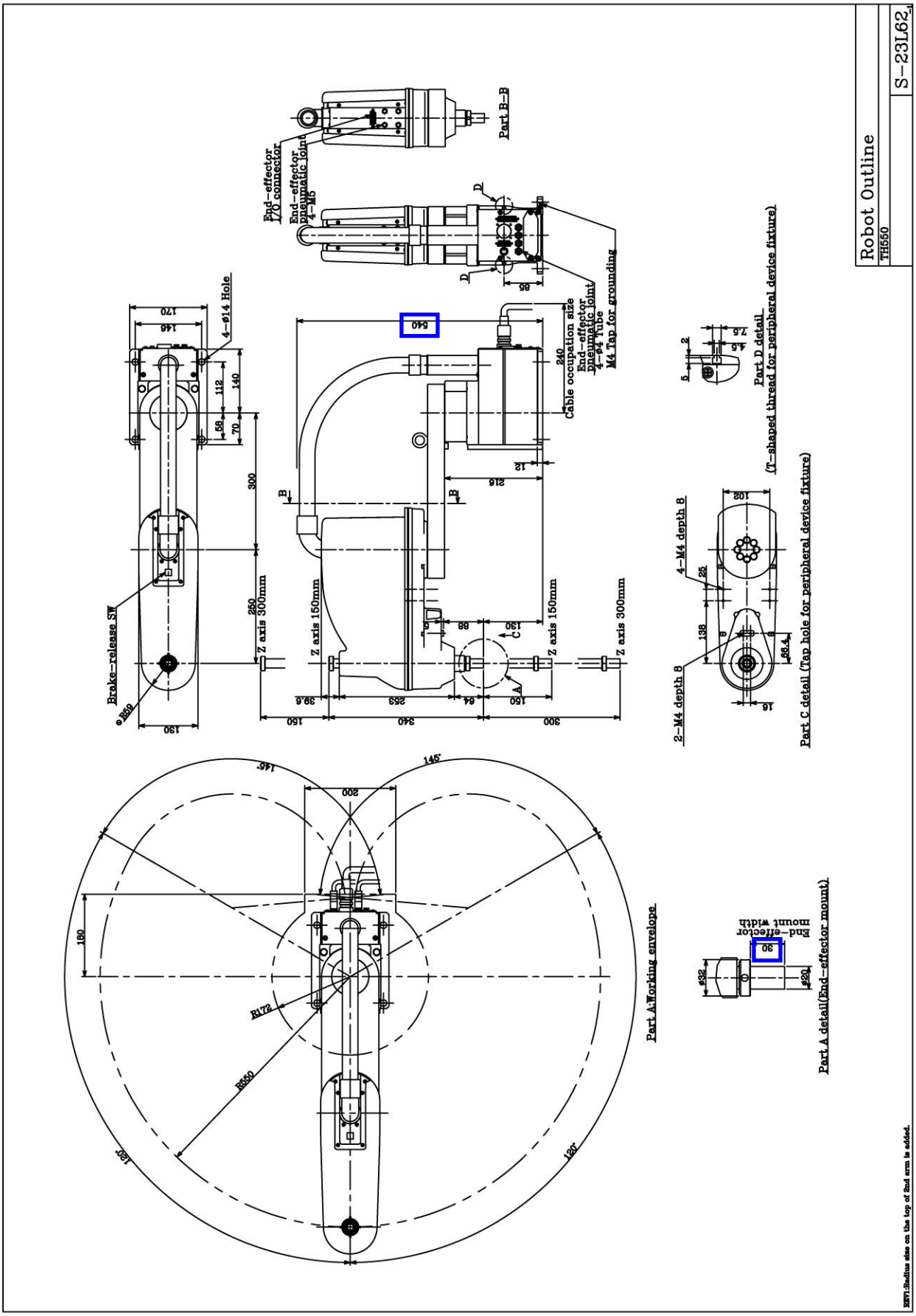
*1 : Maximum speed rates and acceleration rates are limited depending on motion patterns, payload mass, and load inertia.
*2: Continuous operation of standard cycle motion pattern is not possible beyond the effective load ratio.
(Horizontal 300 mm, vertical 25 mm, round-trip, coarse positioning)
*3: When the environment temperature is constant.
*4 : For information about the availability of other options, please contact the Sales Department.

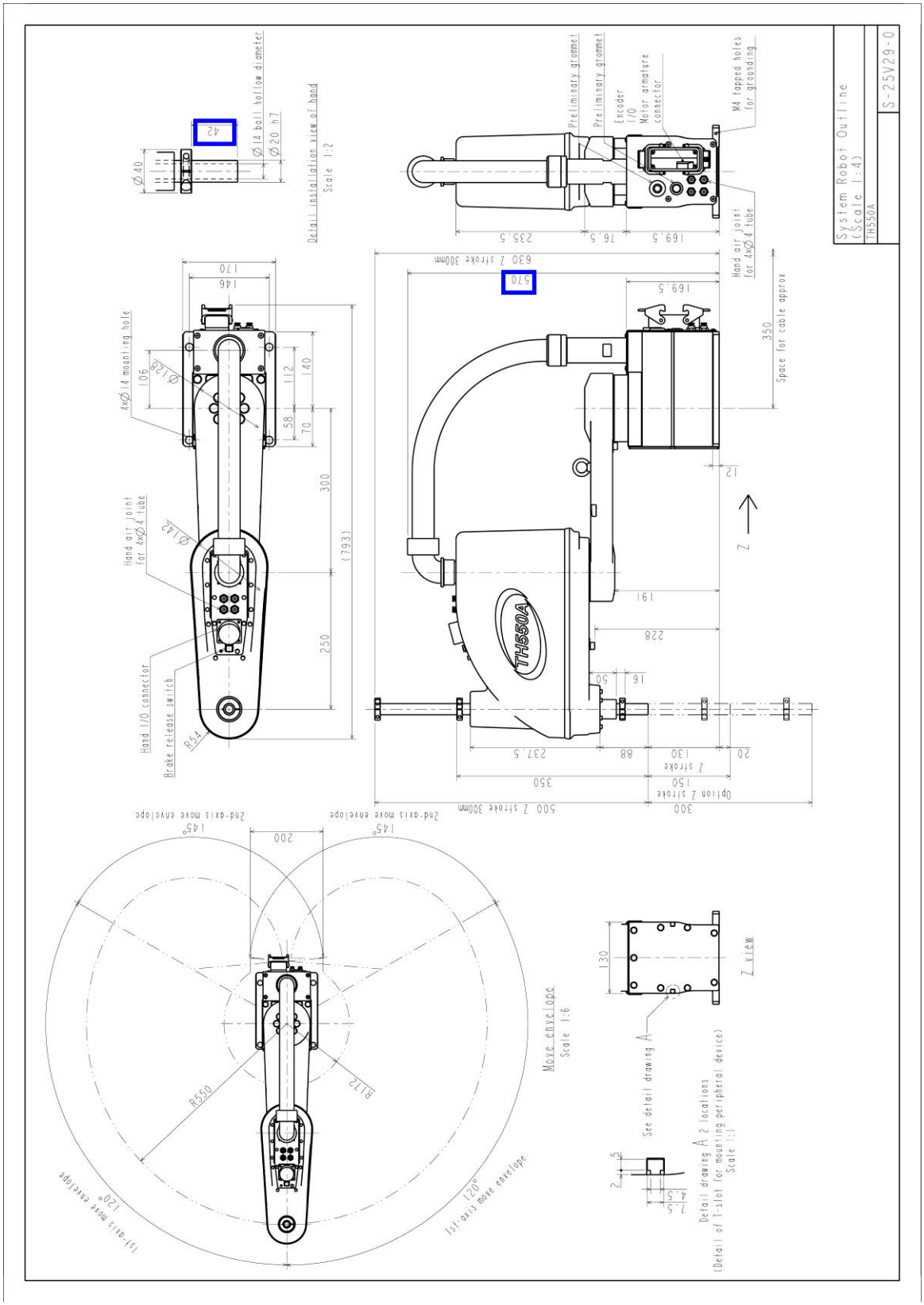
2. Robot Height and Z-axis Position





The TH450 has a total height of 540 mm. The **TH450A** has a total height of 570 mm, which is 30 mm higher than the TH450.
Also, the distance from the bottom surface of the Z axis of the hand mounting area to the stopper is different between these two models (TH450: 30 mm, **TH450A: 52 mm**).


Robot Outline
TH550
S-23L62
Part B-B
End-effector I/O connector
End-effector pneumatic joint 4-M5
End-effector pneumatic joint 4-φ4 Tube
M4 Tap for grounding
Cable occupation size
640
240
216
12
Part D detail (T-shaped thread for peripheral device fixture)
4-φ14 Hole
170
146
112
140
58
70
300
250
Z axis 300mm
Brake-release SW
φ69
180
Z axis 150mm
B
C
A
88
130
39.6
253
64
150
340
300
4-M4 depth 8
2-M4 depth 8
102
26
138
98.4
16
Part C detail (Tap hole for peripheral device fixture)
Part A:Working envelope
200
180
R172
R550
145°
120°
End-effector mount width
30
φ32
φ20
Part A detail(End-effector mount)

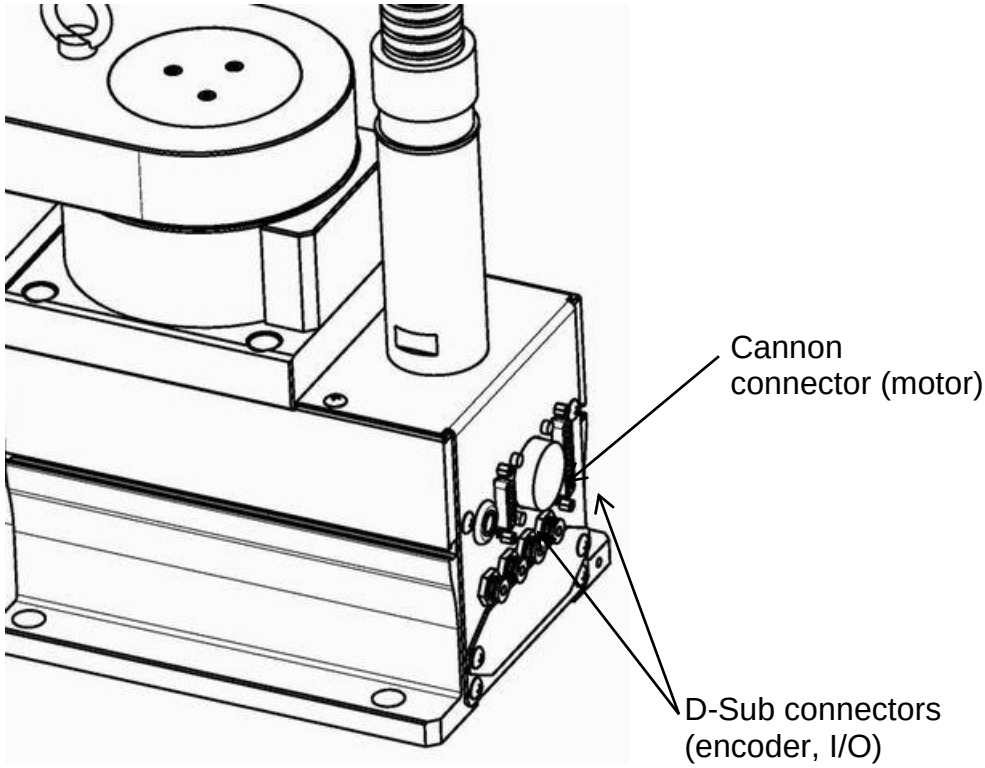


The TH550 has a total height of 540 mm. The **TH550A** has a total height of 570 mm, which is 30 mm higher than the TH550.
Also, the distance from the bottom surface of the Z axis of the hand mounting area to the stopper is different between these two models (TH550: 30 mm, **TH550A: 42 mm**).

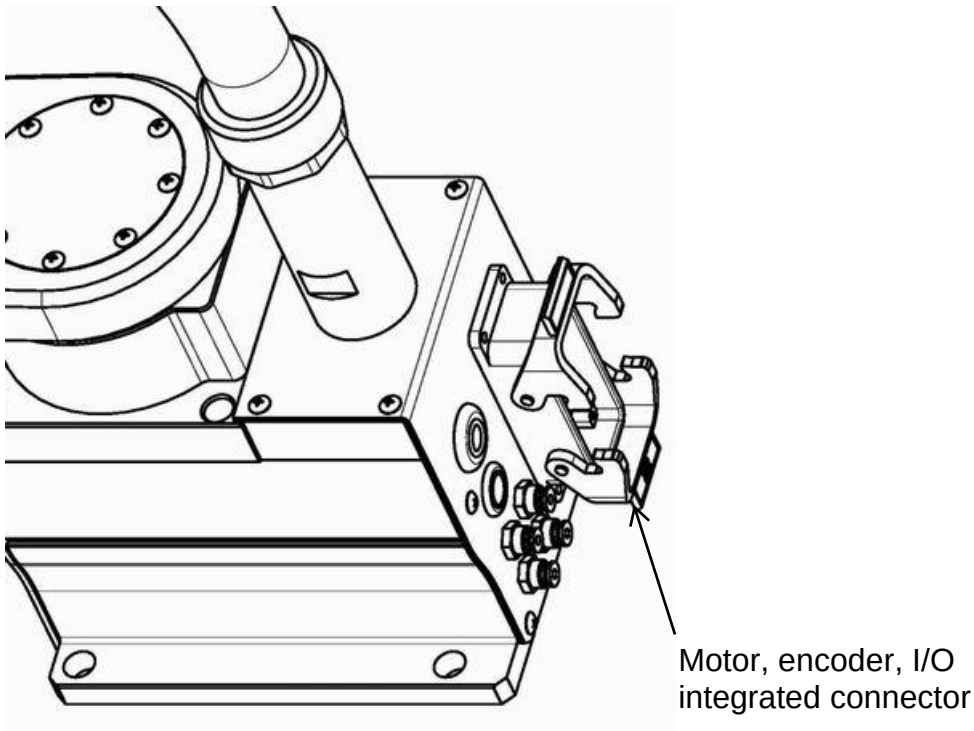
3. Cables Between Robot and Controller

The TH450/TH550 use one Cannon connector and two D-Sub connectors on the robot side.

The **TH450A/TH550A** use a single integrated connector.



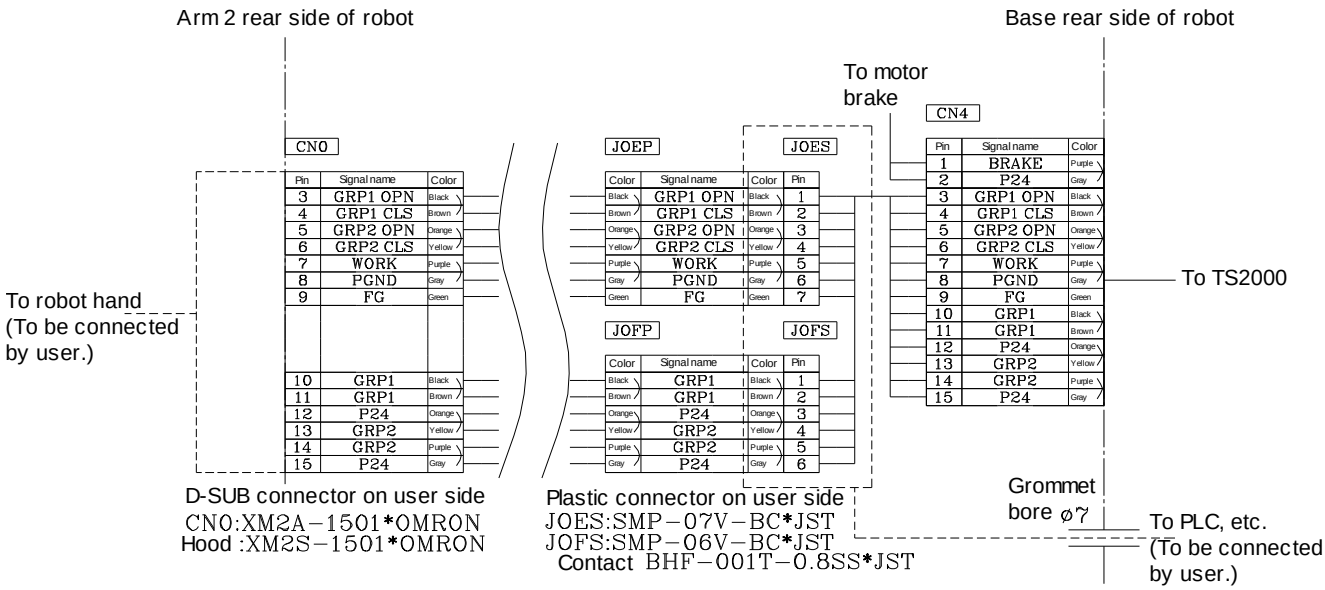
TH450/TH550



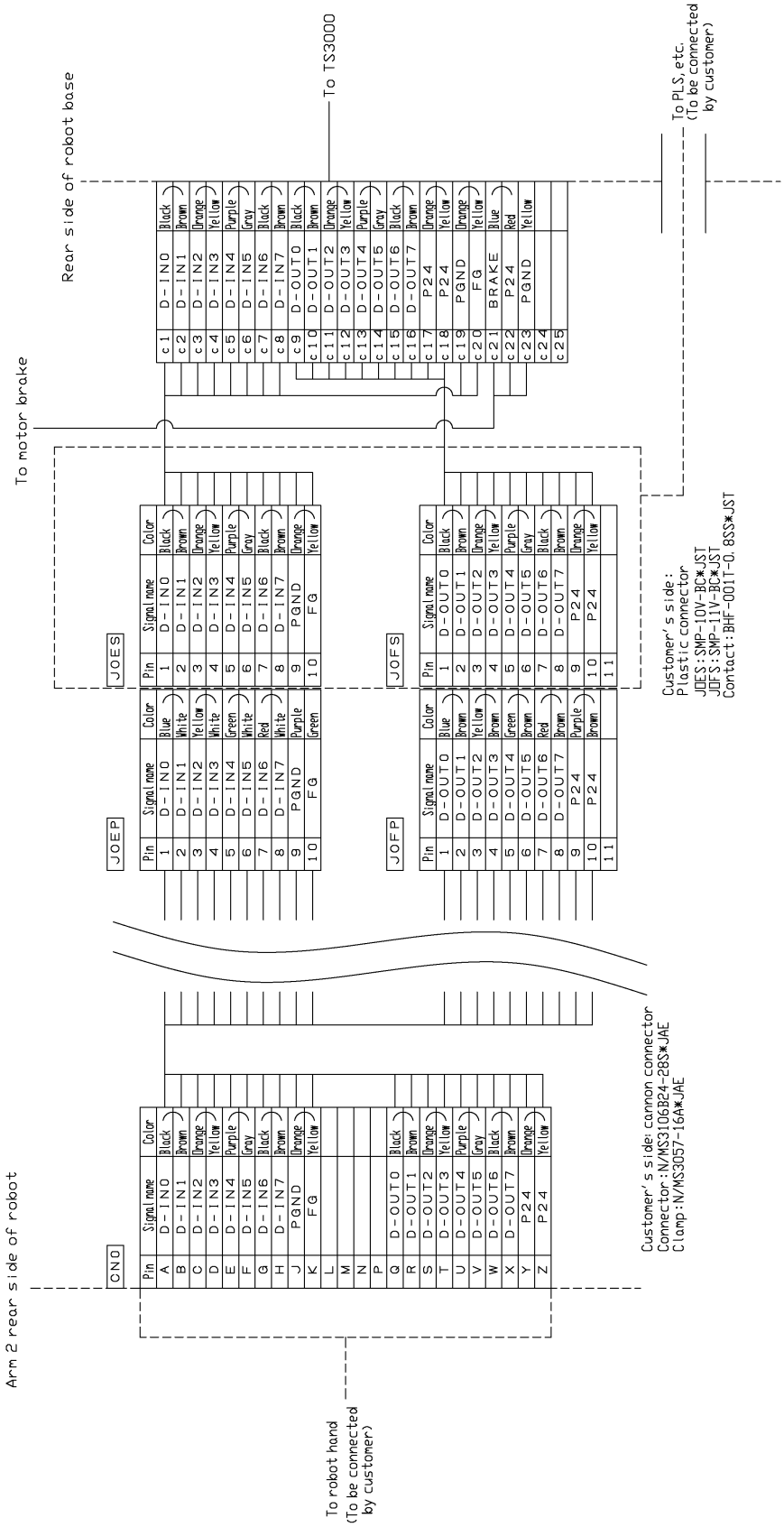
TH450A/TH550A

4. HAND Input Signal Layout and Number of Inputs/Outputs

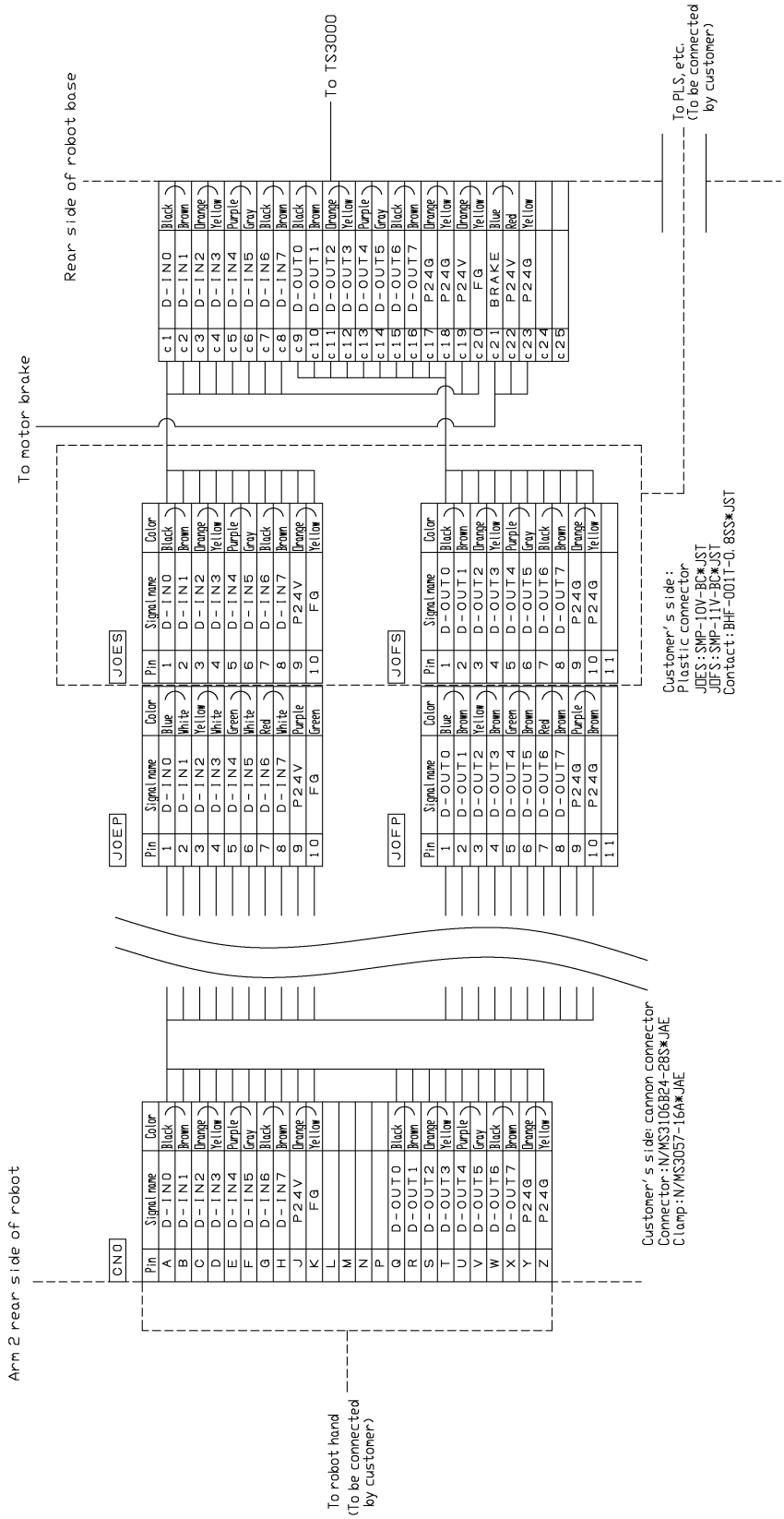
The TH450/TH550 are provided with 5 inputs and 4 outputs for the CN0 (D-Sub) connector.
The **TH450A/TH550A** are provided with 8 inputs and 8 outputs for the CN0 (cannon) connector.
The connector position may vary.



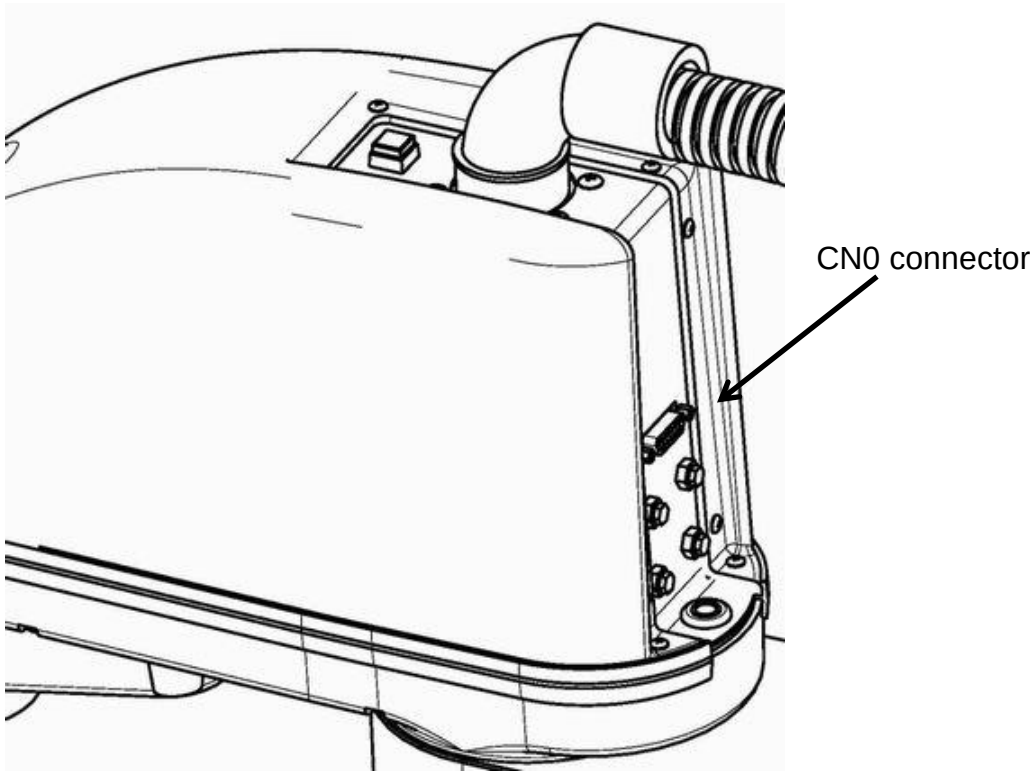
TH450/TH550 Tool Wiring



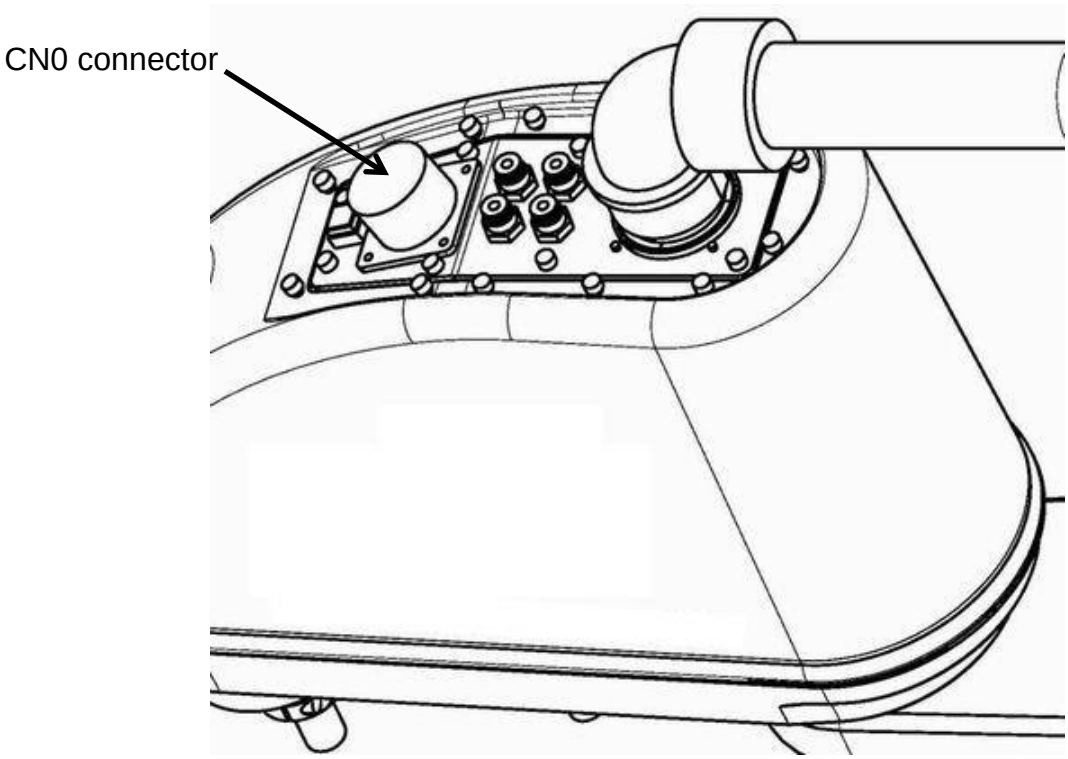
TH450A/TH550A Tool Wiring (Type-N)



TH450A/TH550A Tool Wiring (Type-P)



TH450/TH550



TH450A/TH550A